



BanHaluEval

1 message

ISHMAM TASHDEED, Lecturer, CSE <ishmamtashdeed@iut-dhaka.edu>
To: "Ajwad Abrar Mostofa, Junior Lecturer, CSE" <ajwadabrar@iut-dhaka.edu>

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1. Arithmetic Slip

Keep logic structure intact but insert a subtle numerical error.

Example:

$$60/1.5=45$$

Everything else remains coherent.

This is the simplest form.

2. Formula Misapplication

Use a related but wrong formula.

Example:

Instead of:

$$\text{speed}=\text{time}\text{distance}$$

use:

$$\text{speed}=\text{distance}\times\text{time}$$

Produces:

$$60\times1.5=90$$

Still superficially plausible.

3. Variable Confusion

Swap meanings of variables or quantities.

Example:

"The train traveled 1.5 km in 60 hours."

This often appears in real LLM hallucinations.

4. Invalid Deduction

Inject a logically unsupported transition.

Example:

1. $a > b$
2. $b > c$
3. Therefore $a = c$

This is excellent for symbolic reasoning datasets.

5. Hallucinated Intermediate Fact

Introduce a fabricated assumption.

Example:

"The question implies the train stopped midway."

Then continue consistently.

This mimics realistic reasoning drift.

6. Semantic Drift

Slightly reinterpret the task midway.

Example:

Problem asks for speed, but chain starts computing acceleration.

Very common in LLM reasoning failures.

PROMPT:

You are generating synthetic reasoning hallucinations for a research dataset.

Given:

- a math problem
- a correct reasoning chain
- a correct final answer

Your task:

Modify EXACTLY ONE reasoning step so that the reasoning becomes incorrect, but still appears plausible and coherent to a human reader.

Requirements:

1. Change only ONE reasoning step.
2. The error should be subtle and believable.
3. Do NOT make the reasoning obviously absurd.
4. All later reasoning steps must remain internally consistent with the modified step.
5. Maintain the original reasoning style and tone.
6. The final answer should usually become incorrect.
7. Do not add explanations or commentary.

Possible error types:

- arithmetic slip
- formula misuse
- variable confusion
- invalid logical deduction
- hallucinated assumption
- semantic drift

Return output in JSON format:

```
{  
  "hallucinated_chain": "...",  
  "error_step": ,  
  "error_type": "...",  
  "hallucinated_answer": "..."  
}
```

Problem:
{PROBLEM}

Correct reasoning chain:
{CHAIN}

Correct answer:
{ANSWER}